PHY1203 – General Physics II
Semester 1 – 2024/2025
Electrostatics and Magnetism
Practice Questions

Coulomb's Law

1.

In Fig. 1, two tiny conducting balls of identical mass ***m*** and identical charge ***q*** hang from nonconducting threads of length ***L***. Assume that θ is so small that $\boldsymbol{tan\theta}$ can be replaced by its approximate equal, $\boldsymbol{sin\theta}$.

(a) Show that $x = \left(\frac{q^2 L}{2\pi\varepsilon_0 mg}\right)^{1/3}$ gives the equilibrium separation of x of the balls.

(b) If ***L*** =120 cm, ***m***=10 g, and $x = 5.0\ \text{cm}$, what is |***q***|?

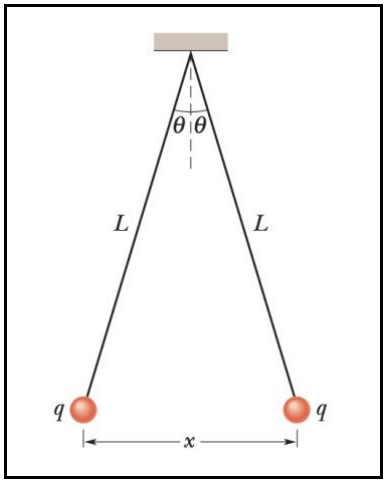


Fig. 1

2.

Fig. 2 shows two concentric rings, of radii R and R′ = 3.00R, that lie on the same plane. Point P lies on the central z axis, at distance D = 2.00R from the center of the rings. The smaller ring has uniformly distributed charge +Q. In terms of Q, what is the uniformly distributed charge on the larger ring if the net electric field at P is zero?

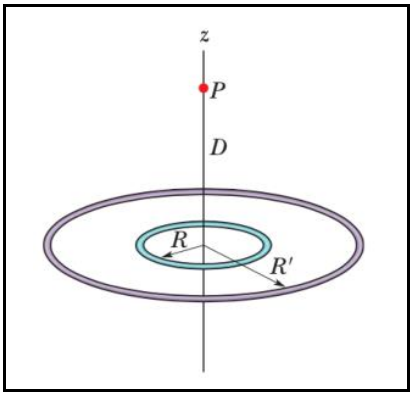


Fig. 2

Gauss Law

3.

A charged particle is suspended at the center of two concentric spherical shells that are very thin and made of nonconducting material. Fig. 3(a) shows a cross section. Fig. 3(b) gives the net flux through a Gaussian sphere centered on the particle, as a function of the radius r of the sphere. The scale of the vertical axis is set by $\Phi_s = 5.0 \times 10^5$ N·m²/C.

(a) Find the charge of the central particle. Find the net charges of (b) shell A and (c) shell B.

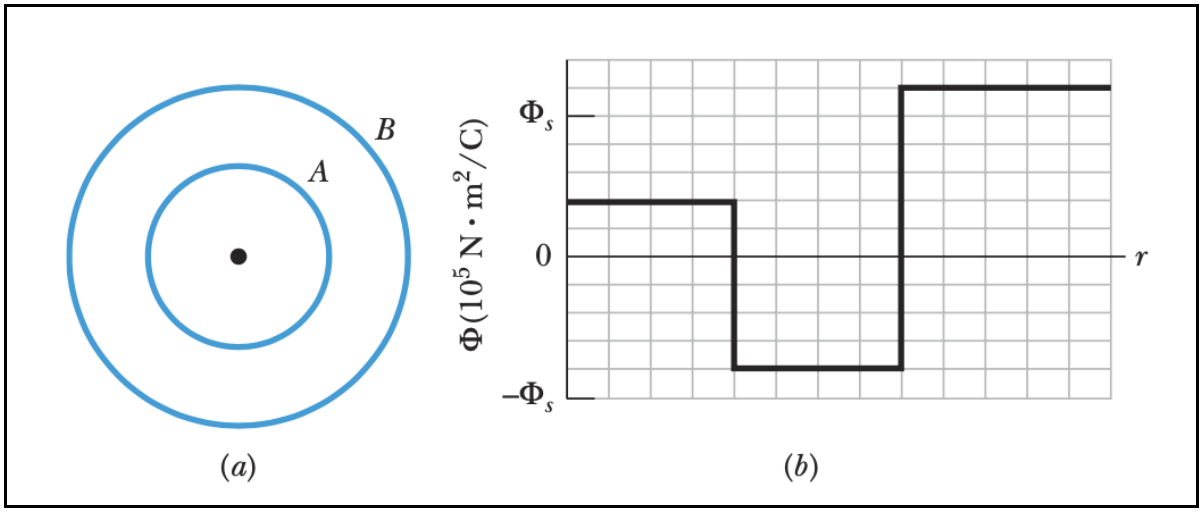


Fig. 3

4.

(a) The drum of a photocopying machine has a length of 42 cm and a diameter of 12 cm. The electric field just above the drum's surface is 2.3×10^5 N/C. What is the total charge on the drum?

(b) The manufacturer wishes to produce a desktop version of the machine. This requires reducing the drum length to 28 cm and the diameter to 8.0 cm. The electric field at the drum surface must not change. What must be the charge on this new drum?

Electric potential

5.

In Fig. 4, what is the net electric potential at the origin due to the circular arc of charge Q_1 = +7.21 pC and the two particles of charges $Q_2 = 4.00Q_1$ and $Q_3 = -2.00Q_1$? The arc's center of curvature is at the origin and its radius is R = 2.00 m; the angle indicated is θ = 20.0°.

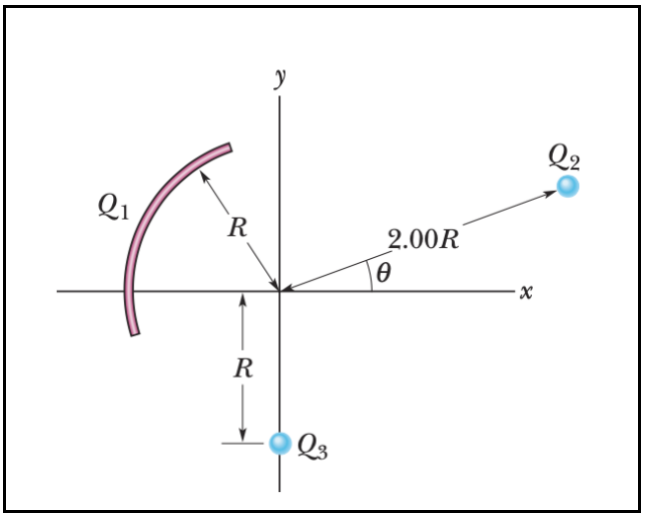


Fig. 4

6.

In Fig. 5, seven charged particles are fixed in place to form a square with an edge length of 4.0 cm. How much work must we do to bring a particle of charge +6e initially at rest from an infinite distance to the center of the square?

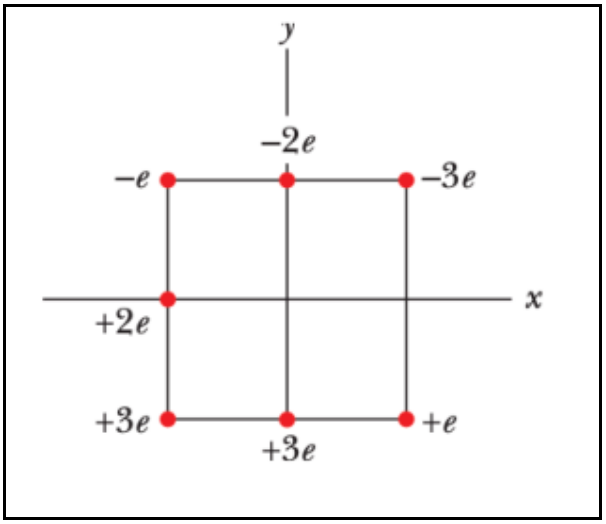


Fig. 5

Capacitance

7.

Fig. 6 shows a parallel-plate capacitor with a plate area A = 5.56 cm² and separation d = 5.56 mm. The left half of the gap is filled with material of dielectric constant κ_1 = 7.00; the right half is filled with material of dielectric constant κ_2 = 12.0. What is the capacitance?

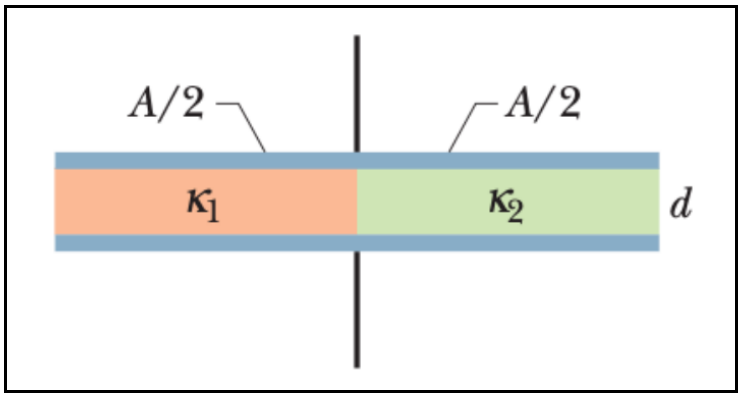


Fig. 6

8.

Fig. 7 shows a parallel-plate capacitor with a plate area A = 7.89 cm² and plate separation d = 4.62 mm. The top half of the gap is filled with material of dielectric constant κ_1 = 11.0; the bottom half is filled with material of dielectric constant κ_2 = 12.0. What is the capacitance?

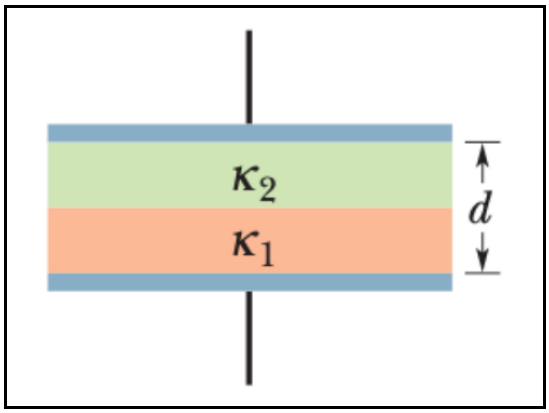


Fig. 7